

Php+mysql

```
<?php
$conn = new mysqli("127.0.0.1", "root", "", "sales_data",
3306);
$query1="SELECT CategoryName, SUM(Revenue) as
total_revenue
FROM sales_data
GROUP BY CategoryName;
";
$store=$conn->query($query1);
while($show=$store->fetch_assoc()){
    echo $show['CategoryName'].
".$show['total_revenue'].<br>";
}
?>
```

Html+php

```
<!DOCTYPE html>

<html>

<head>

    <title>movie complex</title>

</head>

<body>

    <form method="POST" action="index.php">

        Attendees: <input type="text" name="Attendees">

        <button type="submit">calculate</button>

    </form>

</body>

</html>
```

Php

```
<?php

$Attendees=$_POST["Attendees"];

$Seat_Capacity=$_POST["Seat_Capacity"];

$Total_screen=ceil($Attendees/$Seat_Capacity);

echo "Total is".$Total_screen.<br>";

?>
```

Js min

```
let minCT=Math.min(ct1,ct2,ct3);

sql

SELECT * FROM Students
WHERE StudentName LIKE '%im%';
```

Html+js

```
<!DOCTYPE html>
<head>
    <title>Document</title>
</head>
<body>
    calory counter
    <input type="text" id="element-fild">
    <br><br>
    <button id="submit">submit</button>
    <div id="show-output"></div>
    <script src="q1.js"></script>
</body>
</html>
```

Js

```
let arr=[]
let count=0;
document.getElementById("submit").addEventListener("click",function(){
    let clory=document.getElementById("element-fild").value
    clory=parseFloat(clory)

    sum=sum+clory
    arr.push(clory)
    count++;
    //print
    document.getElementById("show-output").innerHTML=`your
array is= ${arr}`
```

sql

```
INSERT INTO book_loans()
VALUES (101,'Abdul','Data Structures',0,0,'Returned'),
(102,'Jabbar','Operating System',12,24,'Overdue'),
(103,'Baekat','Discrete math',5,10,'Overdue'),
(104,'Rahim','Linear Algebra',2,4,'Overdue'),
(205,'karim','Data Structures',15,30,'Lost'),
(106,'Fahim','Operating System',0,0,'Returned')
```

```
$count=0;

$sql= "

SELECT COUNT(*) as total_appoymentment

FROM appointment_records

WHERE DoctorID=402;

";

$result=$conn->query($sql);

if($result->num_rows>0){

    while($row=$result->fetch_assoc()){

        $count=$row["total_appoymentment"];

        echo $row["total_appoymentment"].<br>";

    }

}
```



UIU Web Programming Final Exam - CHEAT SHEET

3 Questions
3 Patterns
Full Marks!

PAGE - 1 (Q 1: JS + Q2: PHP)

PAGE - 2 (Q3: PHP + MySQL)

Q1. JAVASCRIPT (Any Type)

1) HTML Structure

```
<input type="number" id="input">
<button onclick="calculate()">Submit</button>
<p id="result"></p>
<p id="feedback"></p>
<p id="count"></p>
<script src="script.js"></script>
```

Input
Button
Outputs
(Results)

2) JS Structure (Template)

```
let total = 0; // or other variable
let count = 0; // to count attempts/entries
const goal = 2000; // for calorie type
// const secret = Math.floor(Math.random()*4501)+500;

function calculate(){
    let input = Number(document.getElementById("input").value);
    count++; // count increase

    // Update total / score / etc.
    total += input; // or other logic

    // ---- Conditions (Feedback) ----
    if (condition){
        // feedback message
    } else if (condition){
        // feedback message
    } else {
        // feedback message
    }

    // ---- Output ----
    document.getElementById("result").innerHTML = "...";
    document.getElementById("feedback").innerHTML = "...";
    document.getElementById("count").innerHTML = "Entries: "+count;

    // Extra Conditions
    if (count>10 && total<goal){
        document.getElementById("feedback").innerHTML = "Be cautious of frequent snacking!";
    }

    if (count>5 && input !== secret){
        document.getElementById("input").disabled = true;
        document.querySelector("button").disabled = true;
        document.getElementById("feedback").innerHTML = "Out of guesses!";
    }
}
```

3) Common Feedback Conditions (Example)

Calorie Tracker Type

- 0 - 800 → "You're off to a healthy start!"
- 801 - 1600 → "Good progress, keep it balanced!"
- 1601 - 1999 → "Almost at your limit!"
- 2000+ → "Goal reached! Stay mindful!"

Guessing Game Type

- if (guess > secret) → "Too high!"
- else if (guess < secret) → "Too low!"
- else → "Correct!"

Password Strength Type (Scoring)

- Length: +10 for every 2 characters (min 6)
- Uppercase: +15
- Lowercase: +15
- Number: +20
- Special (!@#%&*): +25

Strength Levels

- 0 - 30 → Very Weak
- 31 - 50 → Weak
- 51 - 70 → Medium
- 71 - 90 → Strong
- 91+ → Very Strong
- (100+) → "Perfect Password!"

4) Useful JS DOM

Get Input	Set Output	Disable Input/Button	Generate Random (500 - 5000)
Number(document.getElementById("id").value)	document.getElementById("id").innerHTML = "...";	document.getElementById("id").disabled = true;	Math.floor(Math.random() * (5001-500)+500) // 500 to 5000

Q2. PHP (Calculation Problems)

1) Common Scenario: Pizza / Movie / Venue / Screen / Hall etc.

Input (from HTML)

```
$a = $_POST['a'];
$b = $_POST['b'];
$c = $_POST['c'];
// meaning may change depending on question
```

2) Main Formula

Total/Need to arrange = $a * b$; (or total people)
Need (minimum units) = $\text{ceil}(\text{total} / \text{capacity})$;
Empty/Leftover = $(\text{need} * \text{capacity}) - \text{total}$;
Wasted Money = $\text{empty} * \text{price}$;

3) PHP Function Template

```
function calculate($a, $b, $c){
    $total = $a * $b; // or $total = $a;
    $need = ceil($total/$c);
    $empty = ($need * $c) - $total;
    $waste = $empty * PRICE; // PRICE fixed or taken return [$need, $empty, $waste];
}

// Example calling:
list($need, $empty, $waste) = calculate($a, $b, $c);
```

4) Display Output

```
echo "Total Units Needed: $need <br>";
echo "Empty Seats/Slices: $empty <br>";
echo "Wasted Money (BDT): $waste <br>";
```

5) Classification Type (if needed)

```
if ($total >= 500) $level = "Excellent";
elseif ($total >= 300) $level = "Good";
elseif ($total >= 150) $level = "Average";
else $level = "Poor";
```

Important PHP Functions

ceil() → round up value
floor() → round down
round() → normal round
intval() → integer convert

Comparison (Target)

if (\$total > \$target) → Above target
elseif (\$total < \$target) → Below target
else → Target met exactly
Difference = $\text{abs}(\$total - \$target)$;

Remember: Read question carefully → Identify variables → Apply formula → Show output clearly

1) Database + Table Create + Insert

Create Database & Use

```
CREATE DATABASE campus_library;
USE campus_library;
```

Create Table

```
CREATE TABLE book_loans (
    LoanID INT PRIMARY KEY,
    StudentName VARCHAR(50),
    BookTitle VARCHAR(100),
    DaysOverdue INT,
    PenaltyFee DECIMAL(10,2),
    Status VARCHAR(20)
);
```

Insert Data

```
INSERT INTO book_loans VALUES
(101, "Abdul", "Data Structures", 0, 0.00, "Returned"),
(102, "Jabbar", "Operating Systems", 12, 24.00, "Overdue"),
(103, "Barkat", "Discrete Math", 5, 10.00, "Overdue"),
(104, "Rahim", "Linear Algebra", 2, 4.00, "Overdue"),
(105, "Karim", "Data Structures", 15, 30.00, "Lost"),
(106, "Fahim", "Operating Systems", 0, 0.00, "Returned");
```

2) Connect PHP with MySQL (MySQLi OOP)

```
$conn = new mysqli("localhost", "root", "", "campus_library");
if ($conn->connect_error){
    die("Connection failed: " . $conn->connect_error);
}
```

3) Queries (Four Tasks)

1) Total number of books for each Status (only those having more than 1 entry)

```
$sql = "SELECT Status, COUNT(*) as total_books
FROM book_loans
GROUP BY Status
HAVING COUNT(*) > 1;

// Returned, Overdue (Lost has only 1 so excluded)
```

2) Update: Overdue and DaysOverdue < 7 → Status = 'Grace Period', PenaltyFee = 0

```
$sql = "UPDATE book_loans
SET Status = 'Grace Period', PenaltyFee = 0
WHERE Status = 'Overdue' AND DaysOverdue < 7;
```

3) Update: PenaltyFee > 20.00 Increase by 10% but final fee <= 50.00

```
$sql = "UPDATE book_loans
SET PenaltyFee = PenaltyFee * 1.10
WHERE PenaltyFee > 20.00
AND (PenaltyFee * 1.10) <= 50.00;

// 24 → 26.4 (ok), 30 → 33 (ok)
```

4) Show each BookTitle and total PenaltyFee (Sorted high to low)

```
$sql = "SELECT BookTitle, SUM(PenaltyFee) as TotalPenalty
FROM book_loans
GROUP BY BookTitle
ORDER BY TotalPenalty DESC;
```

4) Display Results in PHP

```
$result = $conn->query($sql);
if ($result->num_rows > 0){
    while ($row = $result->fetch_assoc()){
        echo $row['BookTitle'] . " - " . $row['TotalPenalty'] . "<br>";
    }
} else {
    echo "0 results"
}
```

Use this structure for all SELECT queries

Important SQL Functions

COUNT(*) → Count rows
SUM(column) → Total sum
AVG(column) → Average
GROUP BY column → Grouping
HAVING condition → Filter after grouping
ORDER BY column → Sorting
CASE WHEN THEN → Conditional output

UPDATE Common Syntax

```
UPDATE table_name
SET column = value
WHERE condition;
```

Bonus Update Example

```
UPDATE employees
SET salary = salary + 5000
WHERE salary > 50000
AND salary + 5000 <= 60000;
```

Subquery + CASE (Top Seller Type)

```
SELECT p.ProductName, p.CategoryName,
CASE
    WHEN p.Revenue > (
        SELECT AVG(Revenue)
        FROM sales_data s2
        WHERE s2.CategoryName = p.CategoryName
    ) THEN 'Top Seller'
    ELSE 'Regular Seller'
END as SellerType
FROM sales_data p;
```

Exam Strategy

- ✓ Read the question carefully
- ✓ Identify inputs
- ✓ Apply formula/logic
- ✓ Write clean code
- ✓ Show output clearly
- ✓ Check conditions

ALL THE BEST! 😊